

What is a Protocol?



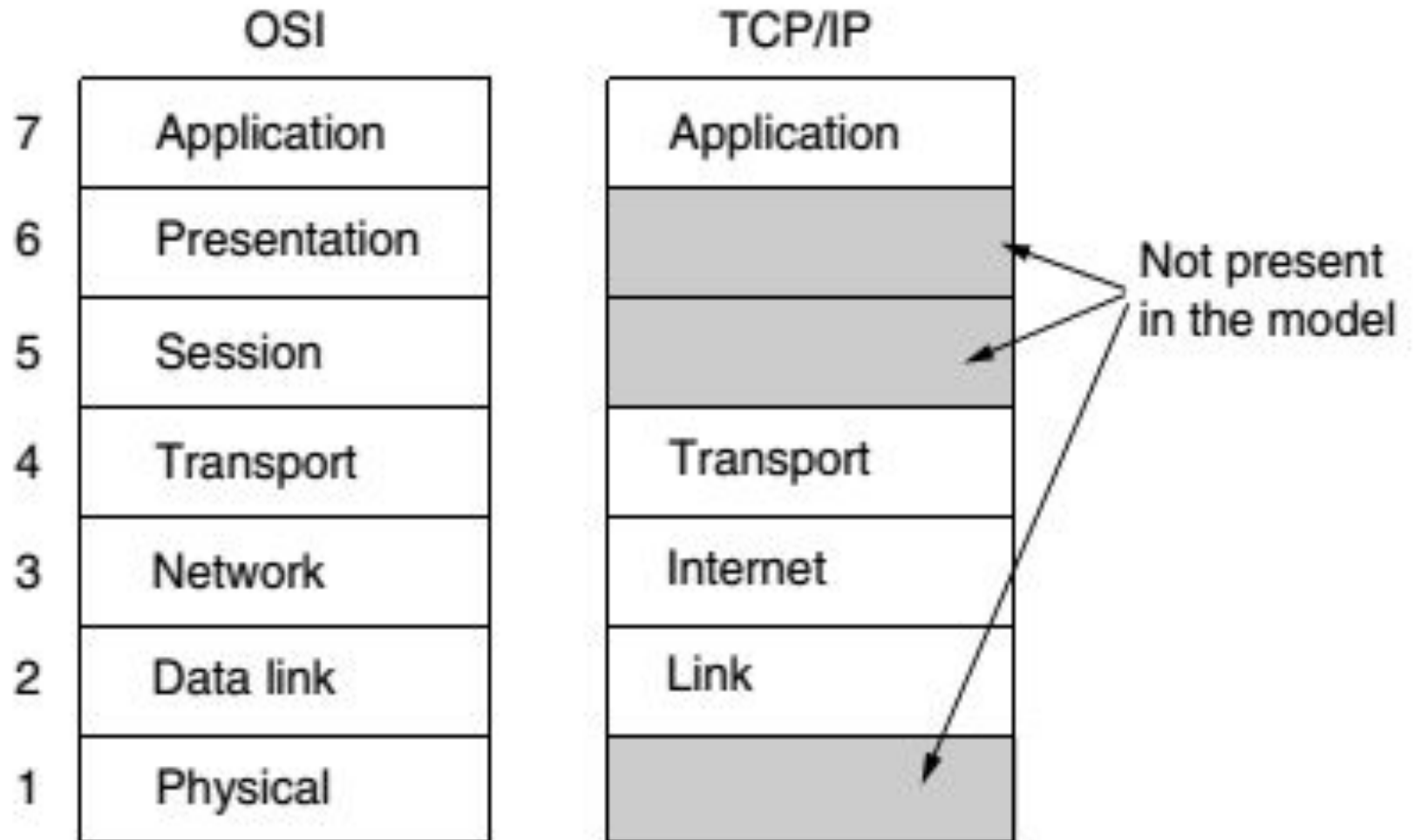
- ◆ Protocol is **set of rules and conventions**.
- ◆ Sender and receiver in data communication must agree on common set of rules before they can communicate with each other.
- ◆ Protocol defines.
 - a) Syntax (what is to be communicated)
 - b) Semantics (how is it to be communicated)
 - c) Timing (When it should be communicated)

TCP/IP Model



- ◆ The TCPIIP protocol suite was developed prior to the OSI model.
- ◆ Therefore, the layers in the TCP/IP protocol suite do not exactly match those in the OSI model.
- ◆ The original TCP/IP protocol suite was defined as having four layers: **host-to-network, internet, transport, and application.**

TCP/ IP Model



TCP/IP Model



- ◆ When TCP/IP is compared to OSI,
- ◆ The **host-to-network layer** is equivalent to the **combination of the physical and data link layers**.
- ◆ The **internet layer** is equivalent to the **network layer**, and
- ◆ the **application layer** doing the job of the **session, presentation, and application** layers.

TCP/IP



- ◆ Transmission control Protocol/Internet Protocol
- ◆ Developed by DARPA
- ◆ No official protocol standard
- ◆ Can identify five layers
 - Application
 - Host-to-Host (transport)
 - Internet
 - Network Access
 - Physical

TCP/IP Network Access Layer

- ◆ Also called as **Host-to-Network** Layer.
- ◆ Performs all functions of physical Layer and Data Link Layer.
- ◆ Exchange of data between end system and network.
- ◆ Address of host and destination
- ◆ Prioritization of transmission.
- ◆ This deals with hardware level, connections as in other network model.

TCP/IP Network Access Layer

- ◆ TCP/IP Protocol Suite includes Host-to-Network Layer protocols such as-
- ◆ **Serial Line Internet Protocol (SLIP) And Point to Point Protocol (PPP).**

TCP/IP Internet Layer



- ◆ An Internet is an interconnection of two or more networks.
- ◆ Internet layer handles tasks similar to network access layer, but between networks rather than between nodes on a network.
- ◆ Uses IP for addressing and routing across networks.
- ◆ Implemented in workstations *and* routers.

TCP/IP Internet Layer



- ◆ This layer is concerned with the format of datagrams as defined in the internet protocol(IP).
- ◆ The protocols in this layer include **Address Resolution Protocol (ARP)**,
- ◆ **Reverse Address Resolution Protocol (RARP)** and
- ◆ **Internet Control Message Protocol (ICMP)**.

TCP/IP Transport Layer



- ◆ This layer is concerned with the transmission of the data.
- ◆ The two main protocols that operate at this layer are **Transmission Control Protocol (TCP)** and **User Datagram Protocol (UDP)**.
- ◆ TCP is reliable transmission protocol and it guarantees that the proper data transfer will take place.
- ◆ UDP is not designed to be reliable or guarantee data delivery.
- ◆ Also called **host-to-host** layer



Functions of Transport Layer

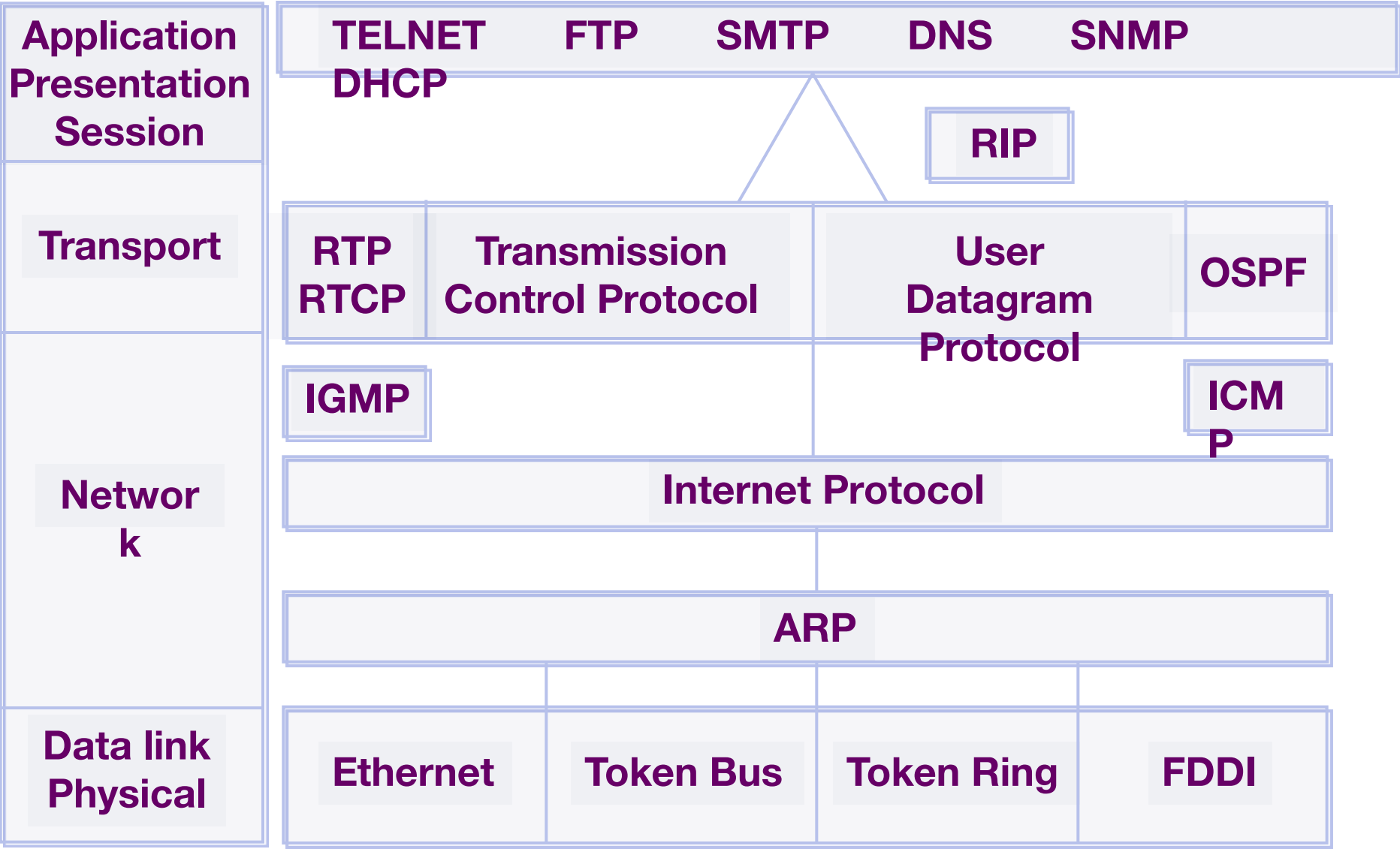
- ◆ **1. Service point addressing:** - Delivery is from specific process on computer to specific process on another computer. For this transport layer uses port addresses.
- ◆ **2. Segmentation and reassemble:** -Each segment of a message contains a sequence number which is used to reassemble the message correctly.
- ◆ **3. Connection control:-**Logical connection is created between source and destination for the duration of complete message transfer.
- ◆ **4. Flow control:-**Flow control is performed end to end.
- ◆ **5. Error control:-**Error control is performed process to process. It ensures that entire message arrives at receivers transport layer without error (damage or loss or duplication). Error correction is done by retransmission.

TCP/IP Application Layer



- ◆ The application layer is concerned with providing network services to applications.
- ◆ There are many application network processes and protocols that work at this layer, including
- ◆ Hyper Text Transfer Protocol (HTTP), Simple Mail Transport Protocol (SMTP) and File Transfer Protocol (FTP).

*TCP/IP



Host-to-Network Layer Protocol

- ◆ Host to network Layer Defines two protocols
 - SLIP
 - PPP
- ◆ SLIP and PPP Protocols allow a user to dial into an ISP over Telephone Line.

SLIP

- ◆ **SLIP-Serial Line Internet Protocol.**
- ◆ It designed to work over **serial ports** and **modem connections**.
- ◆ Defines a sequence of bytes that frame IP packets on a serial line.
- ◆ Commonly used for **point-to-point serial connections** running TCP/IP.
- ◆ It is designed to transmit signals over a serial connection and has **very low overhead**.

SLIP

- ◆ Data transmission with SLIP is **very simple**.
- ◆ This protocol sends a frame composed **only of data to be sent** followed by an end of transmission character (the **END** character, the ASCII code **192**).
- ◆ A SLIP frame looks like:



Problems with SLIP

- ◆ SLIP does not perform **error detection and correction**.
- ◆ SLIP does not provide any **authentication**.
- ◆ SLIP is not approved internet standard.
- ◆ SLIP supports **static IP** address assignment.

PPP

- ◆ PPP means **point to point protocol**.
- ◆ It is a much more developed protocol than SLIP(which is why it is replacing it).
- ◆ It transfers additional data, better suited to data transmission over the Internet.
- ◆ PPP is More Complex than SLIP.

PPP



- ◆ PPP Protocol Supports a set of Authentication Protocol.
- ◆ **Line Control Protocol (LCP):**
 - Responsible for establishing, maintaining, and terminating connection.
- ◆ **Password Authentication Protocol(PAP):**
 - The second is of authentication.
 - Password Authentication is used.

PPP



◆ Network Control Protocol (NCP):

- After authentication is done, PPP sends NCP packet.
- This packet tells ISP server what kind of traffic is to be passed over PPP link.

◆ IP Control Protocol (IPCP):

- Finally the IP packets are exchanged.
- IPCP Establishes and terminates the Network layer connection.

PPP Protocol



- ◆ PPP is point to point protocol.
- ◆ PPP perform error detection.
- ◆ PPP provides authentication and security.
- ◆ PPP is approved internet standard.
- ◆ PPP supports IP and other protocols.
- ◆ PPP supports Dynamic IP address assignment

PPP Frame Format

Flag	1 Byte
Address	1 Byte
Control	1 Byte
Protocol	1 or 2 Bytes
Data and Padding	Variable
Frame check Sequence	2 or 4 Bytes
Flag	1 Byte

Flag: Denotes Start of Frame

Address: Recipient
Control: Control information

Data: Any Data

FCS: for Error Detection

Flag: End of Frame

SLIP Vs PPP

Serial Line Internet Protocol does not establish or maintain connection between the client and ISP server.	In PPP, LCP (Line Control Protocol) is responsible for establishing, maintaining and termination connection between two end points.
Communication starts once the connection between two modems are established.	Communication begins only after authentication and the types of traffic is sent by the client.
Type of traffic cannot be selected in SLIP.	Type of traffic can be selected by NCP
No protocol for termination.	IPCP(IP Control Protocol) terminates a network layer connection between the user and ISP.
No addressing mechanism provided.	Additional services for addressing mechanism is provided
Doesn't allow error control	Allows error control
No provision for data compression	Provides Data compression.

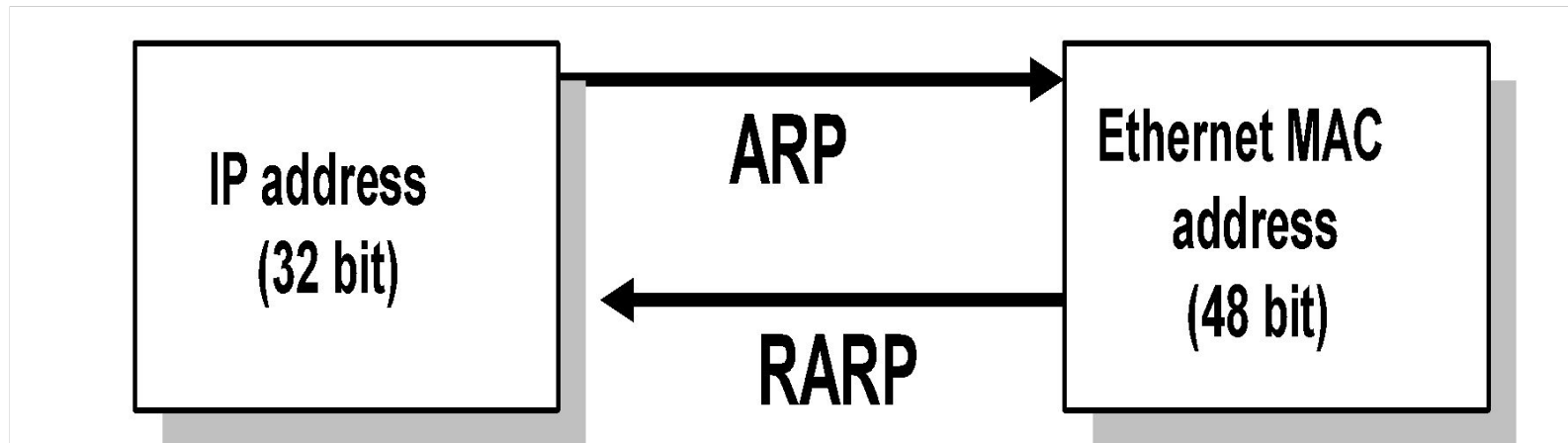
Internet Layer Protocols

- ◆ The Five Network Layer protocols are:
 - ARP
 - RARP
 - IP
 - ICMP

ARP-Address Resolution Protocol

- ◆ ARP takes the IP address of a host as input & gives its **corresponding physical address** as the output.
 - The Internet is based on IP addresses
 - Data link protocols (Ethernet, FDDI, ATM) may have different (MAC) addresses
- ◆ The ARP and RARP protocols perform the **translation between IP addresses and MAC layer addresses**.

ARP And RARP

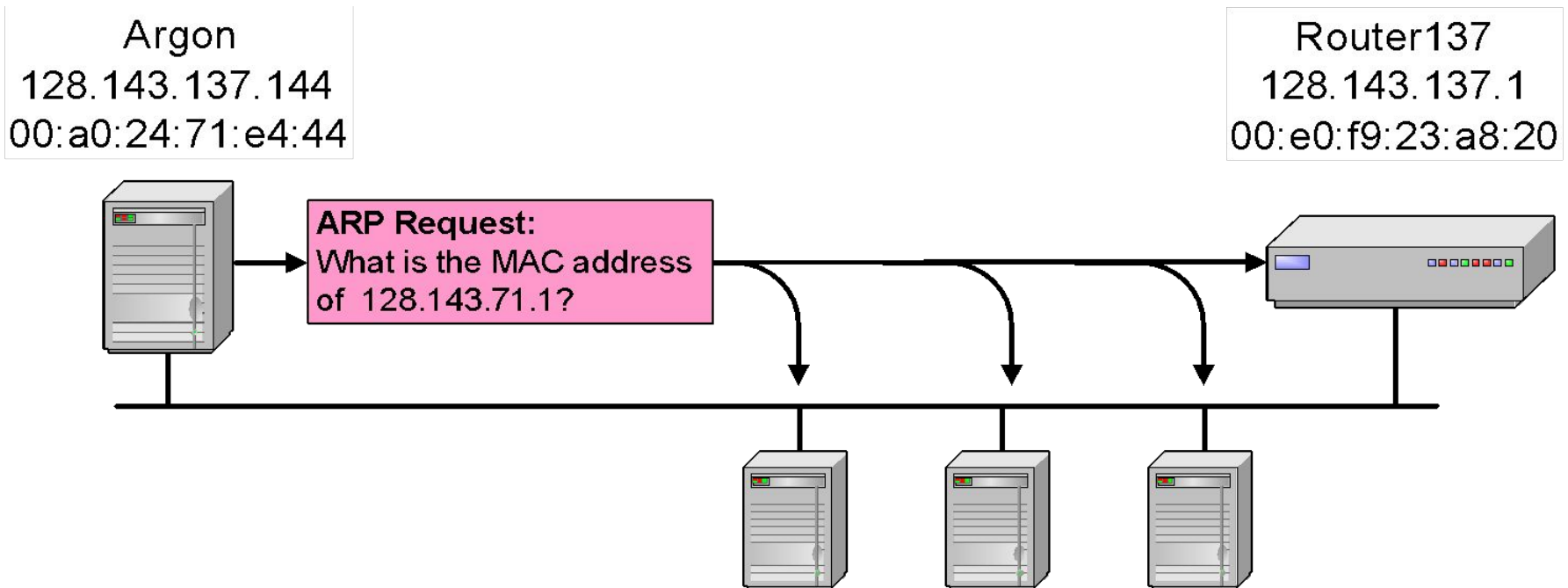


ARP Protocol

- ◆ ARP sends the IP broadcast message to all the computer on the network.
- ◆ The computer whose IP address matches the broadcast IP address sends a reply and along with its physical address to the broadcasting computer.
- ◆ All other computers ignore the broadcast message

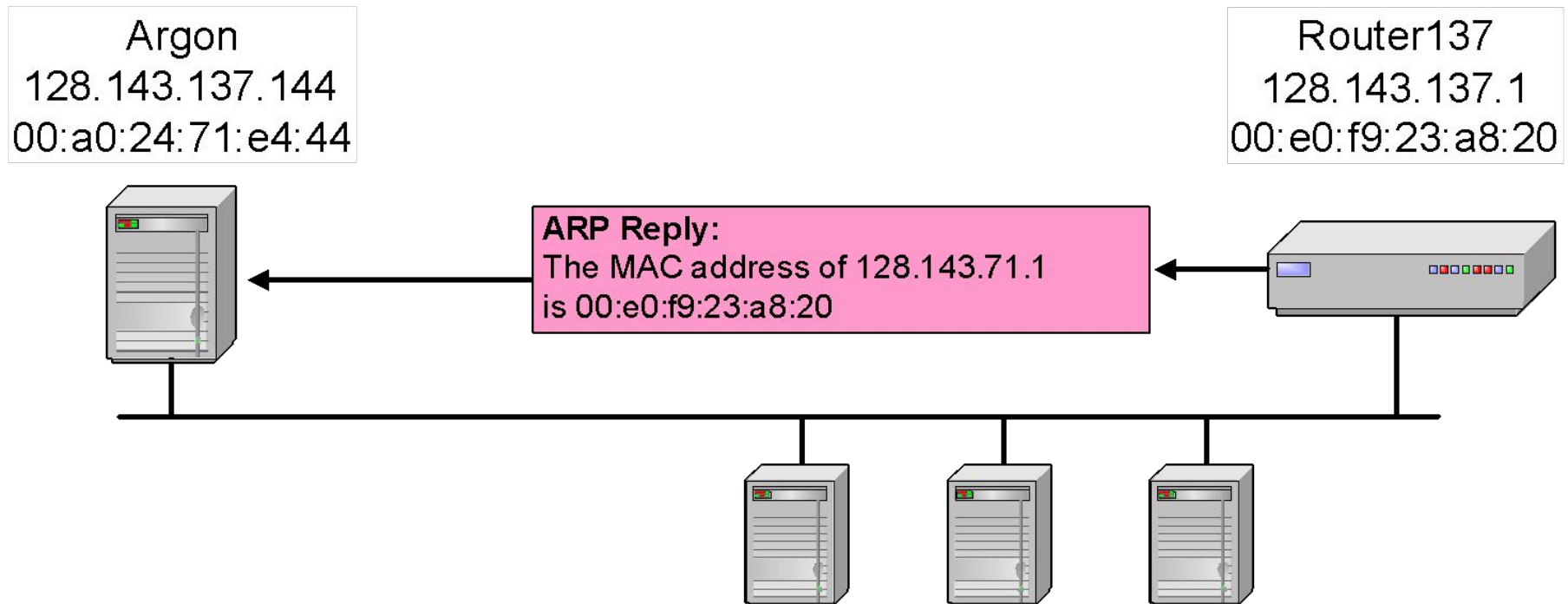
Address Translation with ARP

- ◆ **ARP request:** Argon broadcasts an ARP request to all stations on the network: **“What is the hardware address of Router137?”**

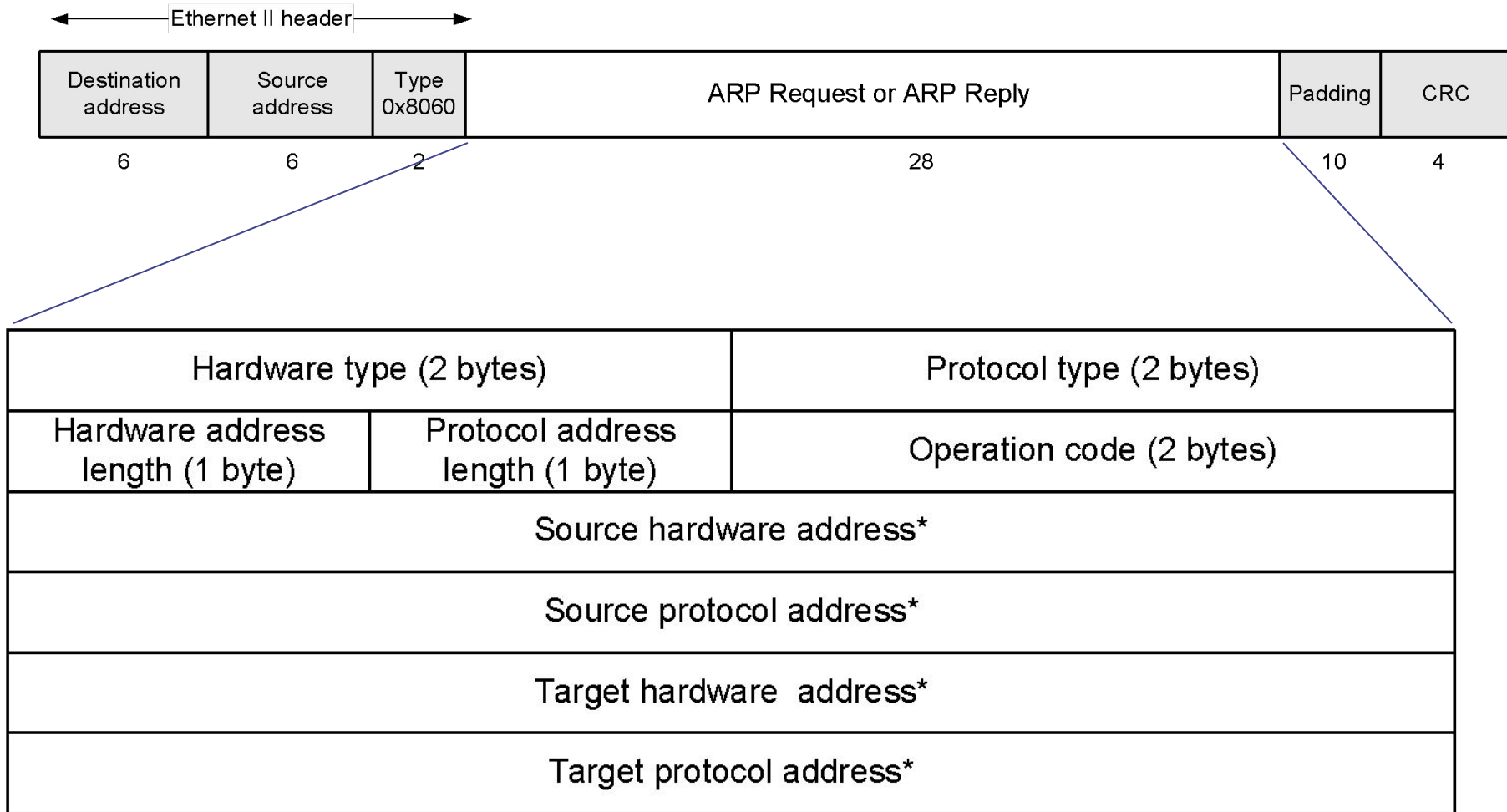


Address Translation with ARP

ARP Reply: Router137 responds with an ARP Reply which contains the hardware address



ARP Packet Format



* Note: The length of the address fields is determined by the corresponding address length fields

RARP

- ◆ Reverse Address Resolution Protocol.
- ◆ If we have to obtain the IP address corresponding to the given Ethernet address.
- ◆ RARP works in very similar way of ARP, but in exactly opposite direction.
- ◆ The RARP server looks at this request.
- ◆ Then it looks up the Ethernet address in its configuration files and sends back the corresponding IP address.

ARP	RARP
ARP converts an Internet Protocol address to its physical network address (MAC).	RARP converts Ethernet MAC address to IP address.
ARP broadcast an IP address in an effort to discover its equivalent hardware address.	RARP broadcast the System's hardware address.
Local host maintain the ARP Table.	A RARP server maintain the RARP table.
RFC 826 describes ARP	RFC 903 describes RARP

Internet Protocol (IP)

- It Is A Routable Protocol Responsible For IP Addressing, Routing, and The Fragmentation and Reassembly Of Packets.
- It Has The Task Of Delivering Packets From The Source Host To The Destination Host Solely Based On The IP Address In The Packet Headers.
- For This Purpose, IP Defines Packet Structures That Encapsulate The Data To Be Delivered.
- It Also Defines Addressing Methods That Are Used To Label The Datagram With Source And Destination Information.

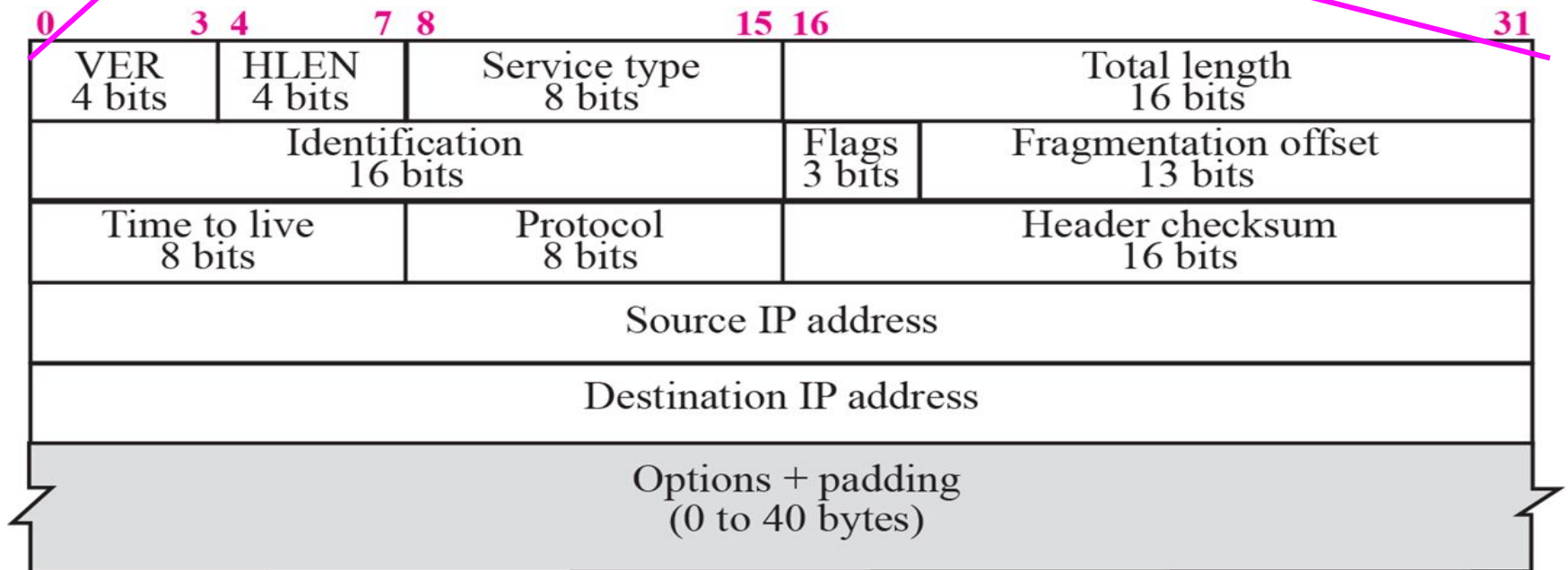
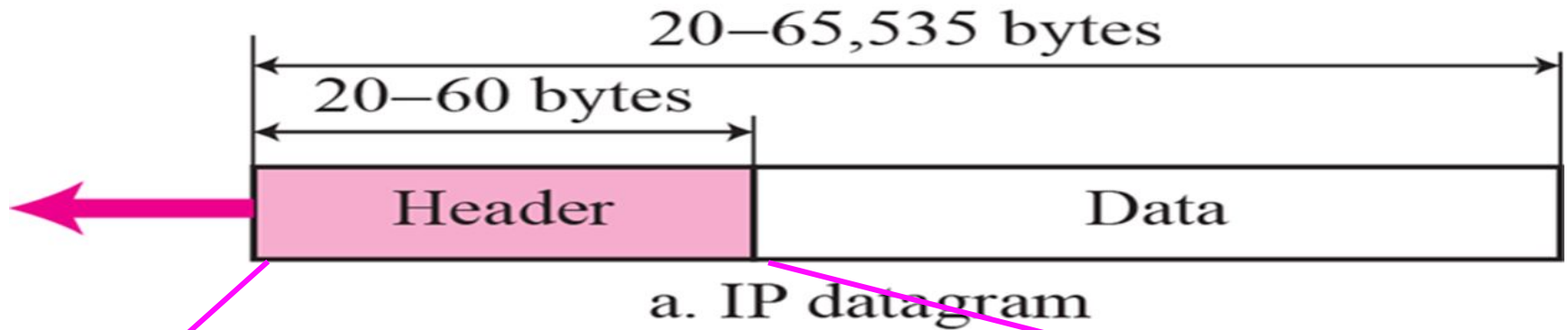
Internet Protocol (IP)

- ◆ IP is internet Protocol.
- ◆ It is **unreliable protocol** because it does not provide any **error control and flow control**.
- ◆ Packets in IP are called “**Datagram**”.
- ◆ Datagram is variable length packet with two parts –header and data



a. IP datagram

IP Header Format



b. Header format

The IPv4 header is comprised of 12 required fields and 1 optional field.

The minimum length of the header is 160 bits (20 bytes).

<i>Field</i>	<i>Length</i>	<i>Description</i>
Version	4 bits	<i>Version of IP (in this case, IPv4)</i>
Internet Header Length	4 bits	<i>Specifies the length of the IP header (minimum 160 bits)</i>
DSCP	8 bits	<i>Classifies traffic for QoS</i>
Total Length	16 bits	<i>Specifies the length of both the header and data payload</i>
Identification	16 bits	<i>Uniquely identifies fragments of a packet</i>
Flags	3 bits	<i>Flags for fragmentation</i>
Fragment Offset	13 bits	<i>Identifies the fragment relative to the start of the packet</i>
Time to Live	8 bits	<i>Limits the lifetime of a packet</i>
Protocol	8 bits	<i>Specifies the next upper layer protocol</i>
Header Checksum	16 bits	<i>Checksum for error checking</i>
Source Address	32 bits	<i>Source IPv4 address</i>
Destination Address	32 bits	<i>Destination IPv4 address</i>
Options	Variable	<i>Optional field for various parameters</i>



Fields of the IP Header

◆ Protocol (1 byte):

- Specifies the higher-layer protocol.

Protocol Number	Upper-Layer Protocol
1	ICMP
2	IGMP
6	TCP
9	IGRP
17	UDP
46	RSVP
47	GRE
50	IPSEC ESP
51	IPSEC AH
88	EIGRP
89	OSPF

◆ Header checksum (2 bytes):

- A simple 16-bit long checksum which is computed for the header of the datagram.
- The receiving host will discard the packet if it fails the checksum calculation.



Functions of the IP

1. Addressing:

- In order to perform the job of delivering datagrams, IP must know where to deliver them to. For this reason, IP includes a mechanism for host addressing.

2. Data Encapsulation and Formatting/ Packaging:

- IP accepts data from the transport layer protocols UDP and TCP.
- It then encapsulates this data into an IP datagram using a special format prior to transmission.

Functions of the IP

3. Fragmentation and Reassembly:

- IP *fragment* IP datagrams **into pieces**.
- The receiving device uses the **reassembly function** to recreate the whole IP datagram again.

4. Routing / Indirect Delivery:

- ◆ When an IP datagram must be sent to a destination on the **same local network**, this is done using ***direct delivery***.
- ◆ However, if the final destination is on a distant network not directly attached to the source datagram must be **delivered *indirectly***.

IPv6



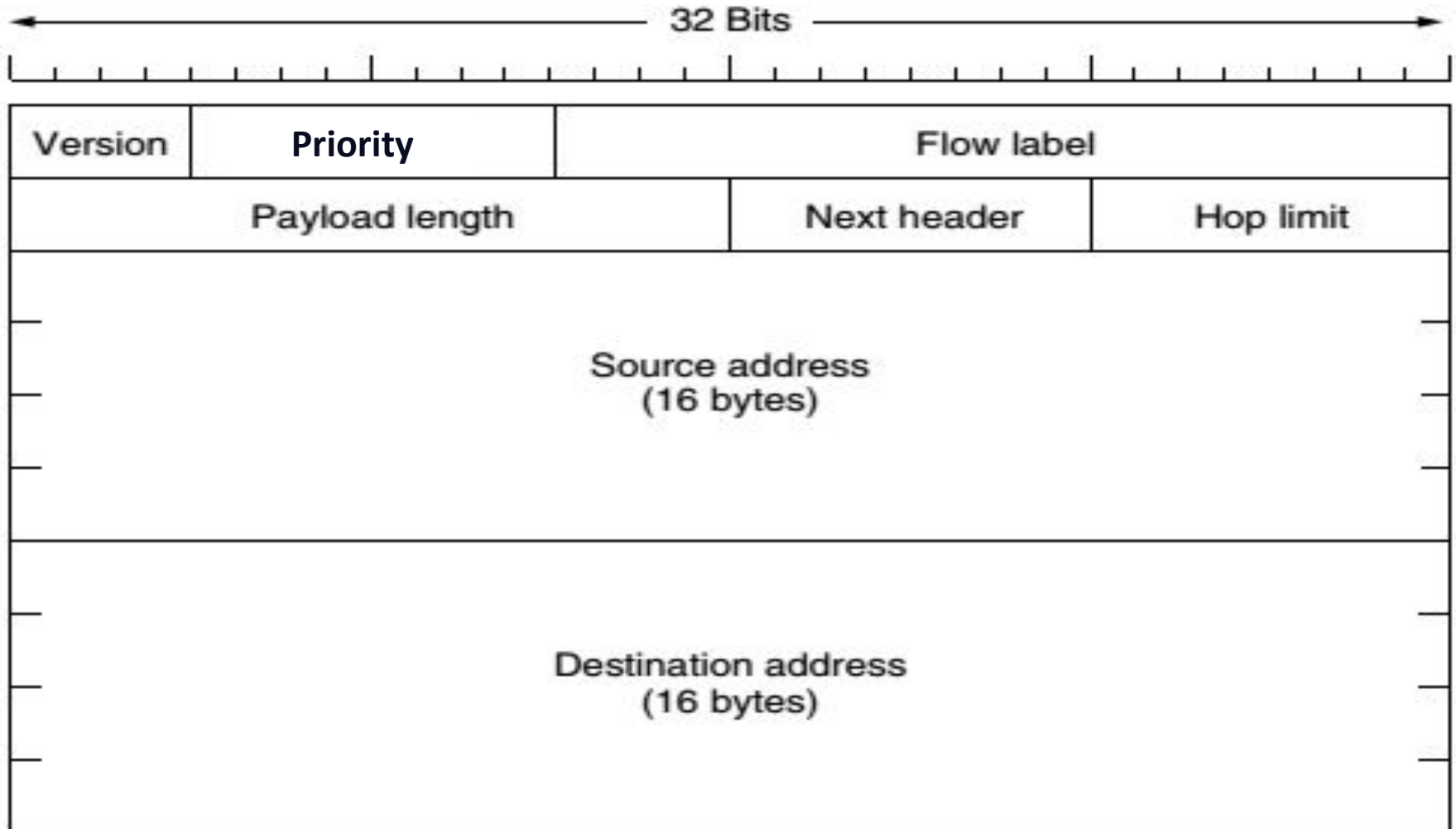
- ◆ IP version 6 (IPv6) is an advanced version of IPv4.
- ◆ It takes all good features of IPv4 and adds new ones.
- ◆ **Larger address space:** IPv6 uses 128 bit(16 Bytes) Address.
- ◆ **Better header format:** This simplifies and speeds up the routing process.
- ◆ **New options.** IPv6 has new options to allow for additional functionalities.
- ◆ **Allowance for extension :** IPv6 is designed to allow the extension of the protocol if required by new technologies or applications.

IPv6 major goals :

1. Support **billions of hosts**.
2. Reduce the size of the routing tables.
3. Simplify the protocol.
4. Provide **better security** (authentication and privacy).
5. More attention to the type of service
6. Aid multicasting by allowing scopes to be specified.
7. Make it possible for a host to roam without changing its address.
8. Allow the protocol to evolve in the future.
9. Permit the old and new protocols to coexist for years.



IPv6 Header Format



IPv6 Header Fields



- ◆ **Version (4-bit):** Defines the version number of the IP. For IPv6, the value is 6.
- ◆ **Priority(4-bit):** Defines the priority of the packet with respect to traffic congestion.
- ◆ **Flow label (3-byte /24-bit):** It is designed to provide special handling flow of data.
- ◆ **Payload length(2-byte):** Defines the length of the IP datagram excluding the base header.
- ◆ **Hop limit (8-bit):** Serves the same purpose as the TTL field in IPv4.

IPv6 Header Fields

◆ Next header (8-bit):

- Defines the header that follows the base header in the datagram.
- The next header is either the base header or an extension header used to encapsulate a packet such as a TCP or UDP segment.
- Note that this field in version 6 is *not* a protocol.

<i>Code</i>	<i>Next Header</i>
0	Hop-by-hop option
2	ICMP
6	TCP
17	UDP
43	Source routing
44	Fragmentation
50	Encrypted security payload
51	Authentication
59	Null (no next header)
60	Destination option

IPv6 Header Fields

◆ Source address.:

- The source address field is a 16-byte (128-bit)
- Internet address that identifies the **original source** of the datagram.

◆ Destination address.

- The destination address field is a 16-byte (128-bit)
- Internet address that usually identifies the **final destination** of the datagram.
- However, if source routing is used, this field contains the address of the next router.

IPv4	IPv6
1. Source and destination addresses are 32 bits (4 bytes) in length.	1. Source and destination addresses are 128 bits(16 bytes) in length.
2. Uses broadcast addresses to send traffic to all nodes on a subnet.	2. There are no IPv6 broadcast addresses. Instead, multicast scoped addresses are used.
3. Fragmentation is supported at Originating hosts and intermediate routers.	3. Fragmentation is not supported at routers.
4.IP header include a checksum.	4.IP header does not include a checksum.
5.IP header includes options.	5. All optional data is moved to IPv6 extension headers.

6.IPsec support is optional	6.IPsec support is required in a full IPv6 implementation.
7. No identification of payload for QoS Handling by routers is present within the IPv4 header.	7. Payload identification for QoS handling By routers is included in theIPv6 header using the Flow Label field.
8. IP address represented in decimal number system	8. IP address is represented in hexadecimal number system
9. ‘.’ used as separator	9. ‘:’ used as separator .

ICMP

- ◆ It is internet control message protocol.
- ◆ It **reports error** and sends **control messages**.
- ◆ Error reporting messages include – **destination unreachable**, **source quench**, **time exceed**, **parameter problem**, **redirection** etc.
- ◆ Query message includes –**echo request** and **reply**, **time stamp request and reply**, router solicitation and advertisement, etc.

Transport Layer Protocols

- ◆ Transport Layer Works on top of Internet Layer.
- ◆ It is concerned with transport of packets from the source to destination.
- ◆ In TCP/IP the transport layer is represented by two Protocols:
 - TCP
 - UDP

Transmission Control Protocol

- ◆ **TCP is transmission control protocol.**
- ◆ **It Provides:**
 - **Connection oriented service**
 - **Reliable service**
 - **Stream delivery service**
 - **Sending and receiving buffers**
 - **Bytes and segments**
 - **Full duplex service**
 - **Process to process communication**

TCP



- ◆ TCP is a **connection oriented protocol**.
- ◆ *Connection oriented* means that a **virtual connection is established** before any user data is transferred.
- ◆ If the connection cannot be established, the user program is **notified**.
- ◆ If the connection is ever **interrupted**, the user program finds out there is a problem.

TCP



- ◆ **TCP is Reliable-**
- ◆ *Reliable* means that every transmission of data is **acknowledged** by the receiver.
- ◆ Reliable does not mean that things don't go wrong, it means that we find out when things go wrong.
- ◆ If the sender does not receive acknowledgement within a specified amount of time, the sender retransmits the data.



TCP

◆ Stream delivery service:

- TCP is a stream oriented protocol.
- It allows the sending and receiving process to obtain as a stream of bytes.
- TCP creates a working environment in such a way that the sending and receiving processes seem to be connected by an **imaginary “tube”** This is called as stream delivery service.

TCP : Flow Control

- ◆ **Sending and receiving buffers:**
- ◆ The sending and receiving process may not **produce and receive data at the same speed.**
- ◆ Hence TCP needs buffers for storage.
- ◆ There are two types of buffers used in each direction:
 - 1) Sending buffer
 - 2) Receiving buffer

TCP



◆ Full duplex service:

- TCP offers full duplex service where the data can flow in both the direction simultaneously.
- The TCP segments are sent both the directions.

◆ Process to process communication:

- The TCP uses port numbers as transport layer addresses.
- Also called as Port to Port communication.

TCP Header



1 byte

1 byte

1 byte

1 byte

Source Port			Destination Port		
Sequence Number					
Acknowledgment Number					
offset	Reser.	Control		Window size	
Checksum			Urgent Pointer		
Options (if any)					
Data					

UDP

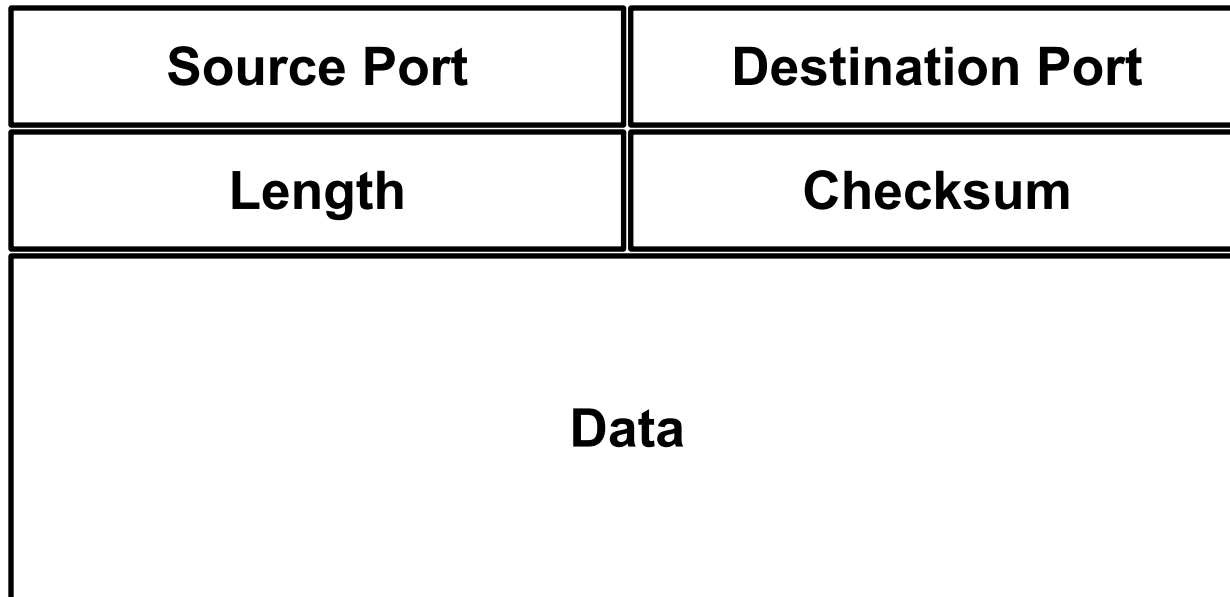


- ◆ UDP is **user datagram protocol**.
- ◆ It is **connectionless** protocol because data is sent without establishing a connection between sender and receiver before sending the data.
- ◆ UDP is **unreliable** because data is delivered without acknowledgement.
- ◆ UDP does not perform Auto retransmission.
- ◆ UDP does **not use flow control**.
- ◆ UDP has **high transmission speed**.

UDP



UDP Datagram Format





TCP	UDP
1. TCP is connection oriented protocol	1. UDP is connection less protocol
2. It provides reliable delivery of messages	2. It provides unreliable delivery of messages
3. It assigns datagram size dynamically for efficiency.	3. Every datagram segment is of the same size.
4. TCP has flow control	4. UDP has no flow control
5. Overhead is low	5. Overhead is very low.
6. Transmission speed is high	6. Transmission speed is very high

	UDP	TCP
Complexity	UDP is less complex	TCP is more complex
Connection	UDP is connection less protocol	TCP is connection oriented protocol
Reliability	It provides unreliable delivery of messages	It provides reliable delivery of messages
Function	By using this protocol one program can send a load of packets to another and that would be the end of the relationship.	As a message makes its way across the internet from one computer to another. This is connection based.
Which layer they exist	Transport layer	Transport layer
Flow controlling	UDP has no flow control	TCP has flow control
Overhead	Overhead is very low	Overhead is low
Which is powerful	UDP is less powerful	TCP is more powerful.

Application Layer Protocols

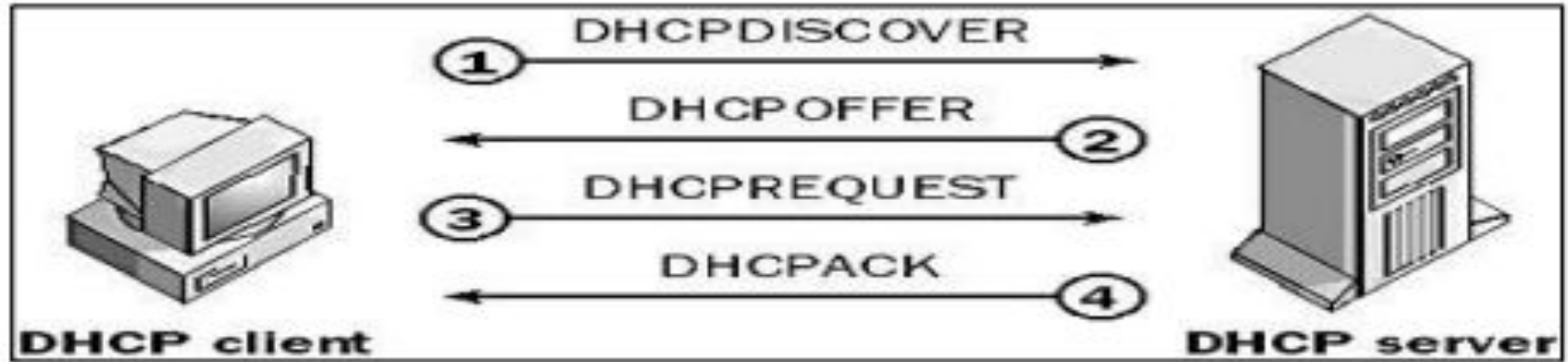
- ◆ DHCP
- ◆ SMTP
- ◆ FTP
- ◆ DNS
- ◆ Telnet

DHCP



- ◆ DHCP (**D**ynamic **H**ost **C**onfiguration **P**rotocol) is a client-server protocol that uses DHCP servers and DHCP clients.
- ◆ A DHCP server is a machine that runs a service that can lease out IP addresses and other TCP/IP information to any client that requests them.
- ◆ The DHCP server typically has a pool of IP addresses that it is allowed to distribute to clients, and these clients lease an IP address from the pool for a specific period of time, usually several days.
- ◆ Once the lease is ready to expire, the client contacts the server to arrange for renewal.
- ◆ DHCP clients are client machines that run special DHCP client software enabling them to communicate with DHCP server.

DHCP



DHCP clients obtain a DHCP lease for an IP address, a subnet mask, and various DHCP options from DHCP servers in a **four-step process**:

DHCP DISCOVER: The client broadcasts a request for a DHCP server.

DHCPOFFER: DHCP servers on the network offer an address to the client.

DHCPREQUEST: The client broadcasts a request to lease an address from one of the offering DHCP servers.

DHCPACK: The DHCP server that the client responds to acknowledges the client, assigns it any configured DHCP options, and updates its DHCP database. The client then initializes and binds its TCP/IP protocol stack and can begin network communication.

DHCP

- ◆ **To configure the DHCP server:**
- ◆ 1. From the Control Panel, go to Administrative Tools >> Computer Management >> Services and Application >> DHCP.
- ◆ 2. From the Action menu, select New Scope.
- ◆ The New Scope wizard is displayed.
- ◆ 3. Enter the following information as prompted:
 - ◆ ☐ Scope name and description:
 - ◆ ☐ IP address range (for example, 192.168.0.170 to 192.168.0.171) ☐ Subnet mask (for example, 255.255.255.0)
 - ◆ Add exclusions (do not exclude any IP addresses)
 - ◆ Lease duration (accept the default of 8 days)
 - ◆ Router (default gateway) of your subnet (for example, 192.168.0.1) ☐
 - ◆ Domain name, WINS server (these are not needed)
 - ◆ Activate Scope? (select “Yes, I want to activate this scope now”)
- ◆ 4. Click Finish to exit the wizard.

DHCP



- ◆ The contents of the DHCP server are listed.
- 5. **Right-click Scope** [iPad dress] scope-name and select Properties.
- 6. In the Scope Properties box, click the **Advanced tab**.
- 7. Select **BOOTP** only, set the lease duration to **Unlimited**, and **click OK**.
- 8. Right-click Reservations.
- ◆ The Controller A Properties box is displayed.
- 9. Enter the IP address and the MAC address for Controller A. **Click Add**. The Controller B Properties box is displayed.
- 10. Enter the IP address and the MAC address for Controller B. Click Add.
- The controllers are added to the right of the Reservations listing.
- 11. Right-click Scope [iPad dress] scope-name to disable the scope.
- 12. Click Yes to confirm disabling of the scope.
- 13. Right-click Scope and select Activate

SMTP



- ◆ SMTP is simple mail transfer protocol.
- ◆ It is **connection oriented** text based protocol.
- ◆ Sender communicates with receiver using a command and supplying data over reliable TCP connection.
- ◆ SMTP is standard application layer protocol for delivery of email over TCP/IP network.
- ◆ SMTP establish a TCP connection between Sender And port number 25 of receiver.

FTP



- ◆ FTP is used for **copying a file** from one host to the other.
- ◆ Some of the problem in transferring files :
 - Two systems may use **different file name** conventions.
 - Two systems may **represent text data in different types**.
 - The **directory structure** of the two systems may be different.
- ◆ FTP provides a simple solution to all these problems.
- ◆ FTP established two connections between the client and server.
- ◆ One is for **data transfer** and the other is for the **control information**.

FTP



◆ Control connection:

- Control connection remains alive during the entire process.
- The IP uses **minimize delay** type services because this is an **interactive connection** between a user and server.

◆ Data Connection:

- Data connection uses the **port 20** at the site.
- This connection is opened when data to be transferred is ready and it is closed when transfer of data is over.
- The service types used by IP is maximize throughput.

TELNET



- ◆ TELNET is abbreviation for **Terminal Network**.
- ◆ It is standard TCP/IP protocol for **virtual terminal services** proposed by ISO.
- ◆ TELNET enables establishment of connection to a remote system in such a way that a local terminal appears to be terminal at remote system.
- ◆ TELNET is general purpose client server application program.

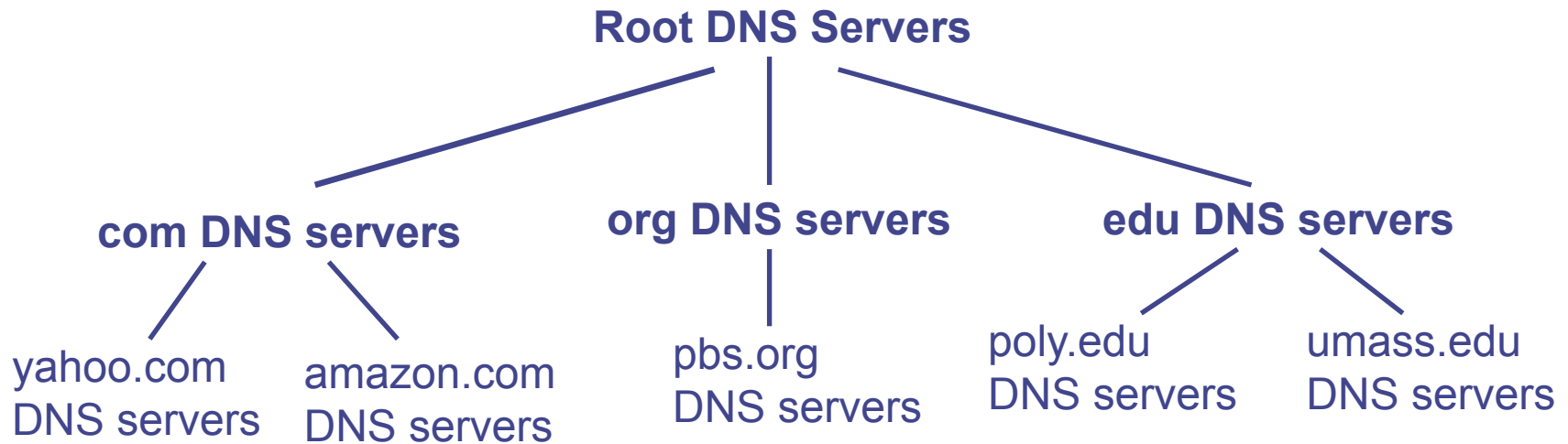
TELNET

- ◆ **Remote login**
- ◆ When user wants to access the application or utility located at the remote machine ,he or she performs remote login.
- ◆ Here the telnet client and server program come into use.
- ◆ The user sends the keystrokes to local operating system. local operating system accept is, but do not interpret them.
- ◆ The characters are send to TELNET client.
- ◆ TELNET client transform the character to a universal character set called Network Virtual Terminal Character and deliver them to the local TCP/IP stack.

DNS-Domain Name System

- ◆ Domain name is human readable name assigned to computer on the internet.
- ◆ Domain refers to group of computers called by common name.
- ◆ DNS is TCP/IP Application that maps Human Readable computer names to IP Addresses.
- ◆ DNS translates internet domain and host names to IP Addresses.

Distributed, Hierarchical Database



Client wants IP for www.amazon.com; 1st approx:

- ◆ client queries a **root server** to find **com** DNS server
- ◆ client queries **com** DNS server to get **amazon.com** DNS server
- ◆ client queries **amazon.com** DNS server to get **IP** address for www.amazon.com

OSI reference model	TCP/IP network model
It has 7 layers	It has 4 layers
Transport layer guarantees delivery of packets	Transport layer does not guarantees delivery of packets
Horizontal approach	Vertical approach
Separate presentation layer	No presentation layer
Separate session layer	No session layer
Network layer provides both connectionless and connection oriented services	Network layer provides only connection less services
It defines the services, interfaces and protocols very clearly	It does not clearly distinguishes between service interface and protocols
The protocol are better hidden and can be easily replaced as the technology changes	It is not easy to replace the protocols
OSI truly is a general model	TCP/IP cannot be used for any other application
It has a problem of protocol filtering into a model	The model does not fit any protocol stack.

◆ **Connectionless protocols:**

- **IP**
- **ICMP**
- **UDP**

◆ **Connection oriented protocol:**

- **TCP**
- **SLIP**
- **PPP**
- **SMTP**